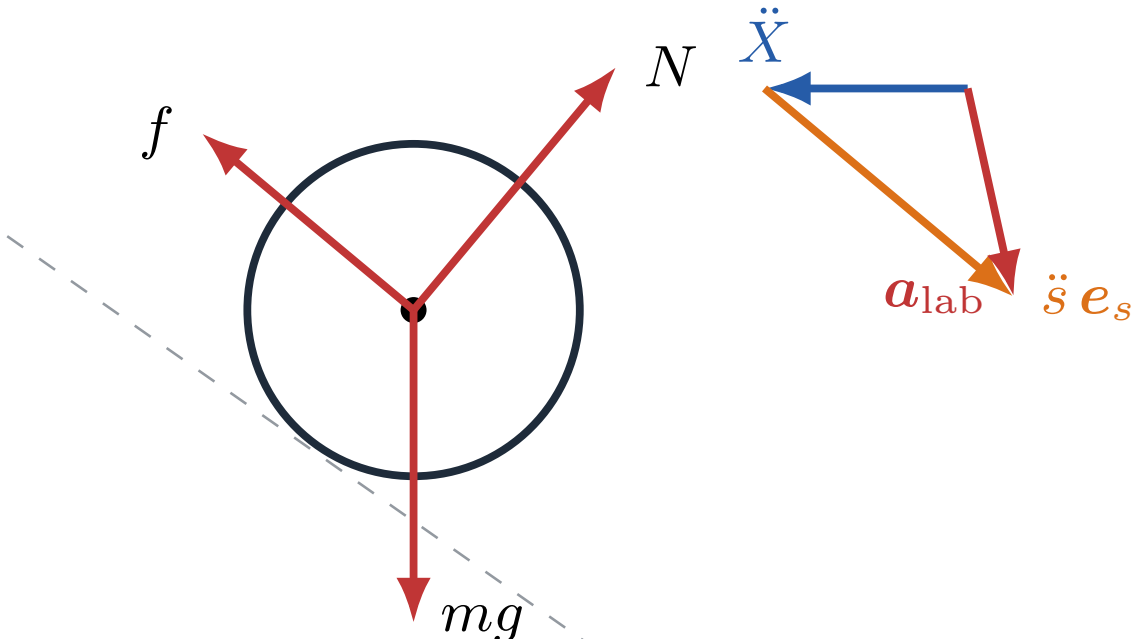

$$\boldsymbol{a}_{\mathrm{lab}} = \ddot{X}\boldsymbol{e}_x + \ddot{s}\boldsymbol{e}_s$$

$$f = \kappa m \ddot{s}$$

$$N = m(g\cos\theta + \ddot{X}\sin\theta)$$